JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 93, NO. B10, PAGES 11,711–11,720, OCTOBER 10, 1988

# Paleomagnetic Constraints on Rotation Within Mount Abbot Quadrangle, Central Sierra Nevada, California

STEPHANIE L. ROSS

*U.S. Geological Survey, Branch of Pacific Marine Geology, Menlo Park, California*
*Department of Geophysics, Stanford University, Stanford, California*

The regional photolineament pattern in Mount Abbot quadrangle in the central Sierra Nevada is sigmoidal in plan view. The fractures in the southwest section of the quadrangle, near Florence Lake, strike N10°–20°E, while those in the central section, along Bear Creek, strike N50°–70°E and those in the northeast strike N25°–40°E. A paleomagnetic investigation was undertaken to determine if this sigmoidal pattern was formed by kinking of originally straight photolineaments after the rock mass cooled below the blocking temperature. If the inner limb of the kink rotated 40°–60° clockwise with respect to the outer limbs, then there should be a similar difference in the paleomagnetic declinations between the Florence Lake and Bear Creek areas. The paleomagnetic results indicate a clockwise difference in declination of 13.4° ± 7° of the Bear Creek area with respect to Florence Lake. Thus the sigmoidal pattern could not have formed entirely by rotation after cooling below the blocking temperature. It is also possible that the observed declination discrepancy is due to tilting, apparent polar wander, or paleomagnetic dispersion instead of rotation. Sigmoidal patterns, formed by kink folding, occur on the outcrop scale with inner limb rotations of 15°–30°; three of these kinks were sampled to test the paleomagnetic method in an area of demonstrable rotation. The data from these sites indicate that although no difference in declination is evident, a reheating event may have occurred which could have removed any rotation from the paleomagnetic record. Reheating, possibly due to the circulation of hydrothermal fluids, may have been causally related to the outcrop scale kinking, since reheating is not observed away from the outcrop scale kinks. The data further indicate that there may have been continuous deformation of the rock mass during the formation of the kinks.

## INTRODUCTION

Although the Sierra Nevada batholith has been regarded as "not internally deformed" in tectonic reconstructions [*Hamilton and Meyers*, 1966], its crystalline rocks are cut by extensive sets of subhorizontal and subvertical fractures. The subvertical sets are recognizable on aerial photographs and are mapped as photolineaments [*Lockwood and Moore*, 1979]. According to *Lockwood and Lydon* [1975] the lineaments obvious in the air photographs are recognized on the ground as linear stream channels, long deep trenches, or parallel lines of trees and shrubs caused by concentrated erosion along closely spaced fractures. *Lockwood and Moore* [1979] show that despite local variations, a regional pattern of northeasterly lineament trends prevails over much of the Sierra Nevada. They note that the trends become more east-northeasterly from north to south and from west to east. *Lockwood and Moore* [1979] also suggest that the broad lineament pattern may reflect variations in regional stresses.

Fracture patterns of Mount Abbot quadrangle (Figure 1) in the central Sierra Nevada have been investigated at both the outcrop and regional scales [*Lockwood and Moore*, 1979; *Segall and Pollard*, 1983*a*, *b*, 1984; *Davies and Pollard*, 1986]. The fractures in the southwest section of the quadrangle, near Florence Lake, strike N10°–20°E and dip steeply eastward [*Segall and Pollard*, 1983*b*]; those in the central section, near Bear Creek, strike N50°–70°E and dip steeply to the southeast [*Segall and Pollard*, 1983*a*]. The fractures in the northeast section strike N25°–40°E, completing a regional scale sigmoidal pattern (Figure 2). At Florence Lake most of the fractures are joints; at Bear Creek the fractures include both faults and joints. The faults formed by slip on older joints [*Segall and Pollard*, 1983*a*]. In all areas the fractures are filled with hydrothermal minerals, mainly epidote and chlorite. Sigmoidal fracture patterns also occur at outcrop scale in the Bear Creek area [*Segall and Pollard*, 1983*a*; *Davies and Pollard*, 1986], as is shown in Figure 3. At this scale the sigmoidal pattern clearly results from the buckling of originally straight slabs of granodiorite, a process of kink folding [*Reches and Johnson*, 1976; *Collier*, 1978]. This kinking was apparently contemporaneous with the faulting.

*Davies and Pollard* [1986] suggest that one tectonic event with a single direction of maximum regional stress might explain the development of both the quadrangle scale sigmoidal pattern of photolineaments and the outcrop scale kink bands. They propose that the fracture strikes were originally the same in the Bear Creek and Florence Lake areas and that the Bear Creek area has rotated 40°–60° clockwise with respect to Florence Lake. An alternative possibility is that the regional sigmoidal pattern is not a kink band and that there has been no rotation of the Bear Creek area with respect to the Florence Lake area. If this is the case, then at a regional scale the fractures originally formed in a sigmoidal pattern.

If this proposed rotation occurred, then the magnetic directions of the rocks in the two areas should show a 40°–60° difference in declination. Hence to test whether or not rotation has occurred since the present magnetization was acquired, oriented samples for paleomagnetic study were collected from both the Florence Lake and Bear Creek areas. Since there is geologic evidence that the outcrop scale kinks did form by rotation, three outcrop scale kinks near Bear Creek were also sampled as a control experiment.

Copyright 1988 by the American Geophysical Union.

Paper number 7B3026
0148-0227/88/007B-3026$05.00

## GEOLOGIC SETTING

The study area is in the central part of the Sierra Nevada batholith which is predominantly composed of granitoids

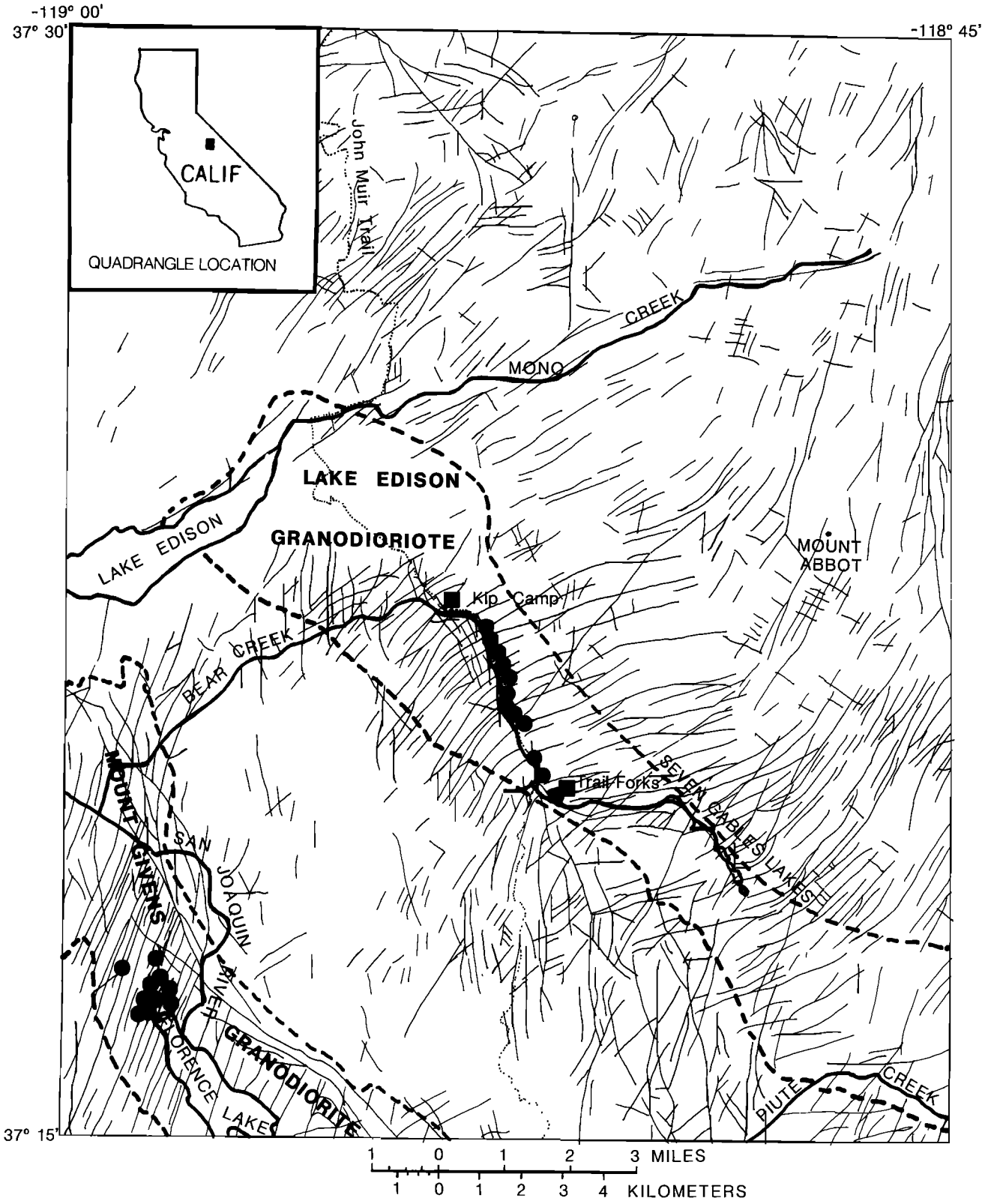



Fig. 1. Map of Mount Abbot quadrangle showing traces of lineaments [after *Lockwood and Lydon*, 1975] and site locations (dots). Also shown are the locations of the Kip Camp and Trail Forks outcrops (squares).

ranging in age from 220 to 80 Ma [*Kistler et al.*, 1965; *Evernden and Kistler*, 1970; *Chen and Moore*, 1982]. The rocks sampled in this study were either granodiorites or fine-grained xenoliths from within the granodiorites. The Florence Lake samples are of the Mount Givens Granodiorite, a medium-grained equigranular biotite-hornblende granodiorite characterized by subhedral to euhedral mafic minerals and minor sphene [*Lockwood and Lydon*, 1975]. It has been dated by the uranium-lead method using zircons at about $90 \pm 3$ Ma [*Stern et al.*, 1981]. The xenoliths contain abundant fine- to

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

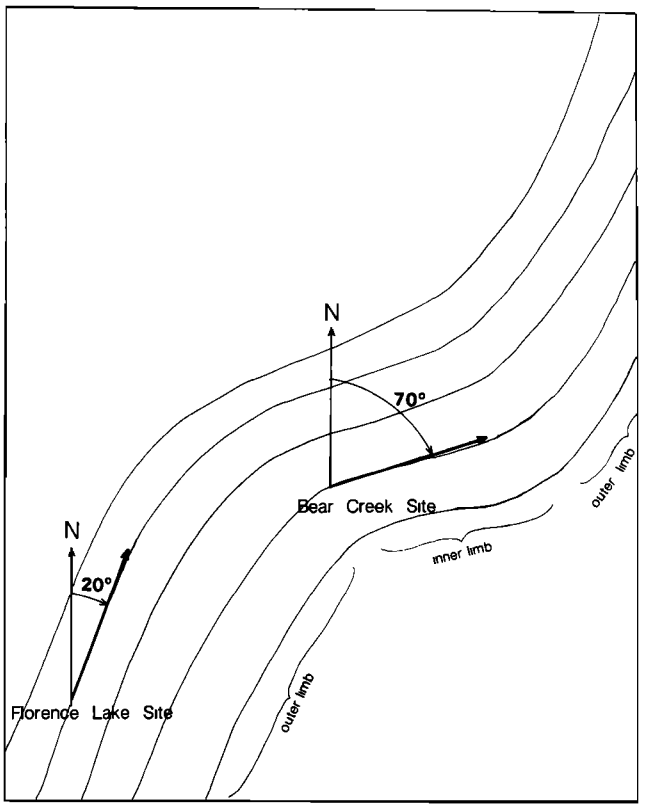



Fig. 2. Schematic of the lineaments in the Mount Abbot quadrangle emphasizing the sigmoidal fracture pattern and defining the terms inner and outer limbs.

medium-grained ferromagnesian minerals. The Bear Creek samples are granodiorite of Lake Edison, a fine- to medium-grained homogeneous biotite-hornblende granodiorite characterized by subhedral mafic minerals and sphene [*Lockwood and Lydon*, 1975]. Uranium-lead dating of zircon from the grandiorite of Lake Edison indicates an age of 90–88 Ma [*Stern et al.*, 1981].

These granodiorite ages provide an upper limit for the ages of the steeply dipping fractures in this area. The exact ages of the jointing and faulting are uncertain [*Segall and Pollard*, 1983*a*]. Muscovites from fault fillings at the Trail Forks outcrop along Bear Creek have been dated by the potassium-argon method at approximately 71 and 75 Ma (S. Martel, personal communication 1987); the fractures must be older than this.

Many of the joints and faults in this area are filled with hydrothermal minerals. A. Kilinc's (unpublished data, 1983) work on the thermodynamics of the joint mineralization at Florence Lake indicates that the hydrothermal fluids were at a temperature of 360°C. A similar mineral assemblage is found in the cracks that formed during the youngest faulting and kinking deformation event in the Bear Creek area. This suggests that the formation of these mineral assemblages may have caused a reheating event that should be considered.

## FIELD AND LABORATORY TECHNIQUES

Seventy-three cores from 12 sites were drilled in the granodiorite and small mafic inclusions of the Lake Edison pluton along Bear Creek (Figure 1 and Table 1). Sixty cores were drilled in 10 sites of Mount Givens Granodiorite and its mafic inclusions, near Florence Lake (Figure 1 and Table 1). The diameters of the sampled inclusions ranged from 6 to 60 cm. All cores were oriented in the field with a magnetic compass and were checked by sun compass, weather permitting, or by back azimuths; the estimated accuracy of orientation is approximately 2°.

In addition, three outcrop scale kinks were sampled near Bear Creek, one at Kip Camp and two at Trail Forks (Figures 1 and 3). Three samples were drilled in each of four or five locations throughout each kink.

One or two specimens were prepared from each core. For 13 of the Lake Edison and six of the Mount Givens samples, specimen pairs were selected, one specimen for thermal and one for alternating magnetic field (af) demagnetization. These 19 samples were selected from 14 sites. One set of specimens was demagnetized using progressive alternating fields ranging from 5 to 80 mT. The second set of specimens was stepwise thermally demagnetized using a Schonstedt TSD-1 shielded furnace. The remanent magnetizations were measured on a cryogenic magnetometer. Although noisy, the vector compo-

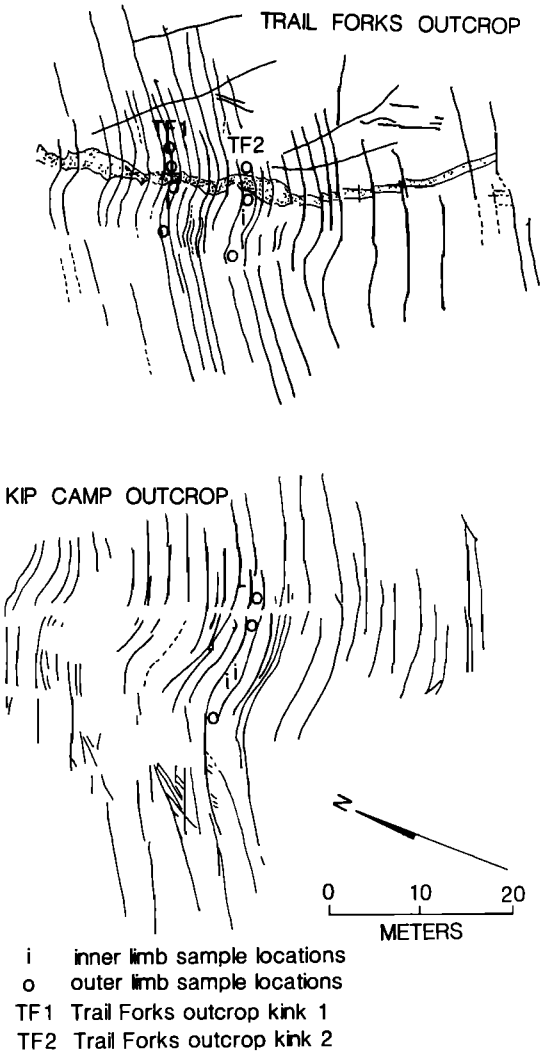



Fig. 3. Map of Trail Forks and Kip Camp outcrops showing fractures [after *Segall and Pollard*, 1983*a*] and sampling locations. Each sampling location marked represents three samples which usually were separated by fractures too numerous to include on this map scale.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Paleomagnetic Results: Bear Creek and Florence Lake

| Site | Type | In Situ Dec, deg | In Situ Inc, deg | Corrected Dec, deg | Corrected Inc, deg | $N$ | $R$ | $\kappa$ | $\alpha_{95}$ | VGP °N | VGP °E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | *Bear Creek* | | | | | | | |
| BC001 | GD-TH | 352.4 | 77.4 | 348.0 | 77.2 | 4 | 3.982 | 165.5 | 7.2 | 60.9 | 230.9 |
| BC007 | GD-TH | 2.6 | 71.4 | 359.6 | 71.4 | 4 | 3.957 | 223.0 | 6.2 | 71.3 | 240.4 |
| BC027A | IN-AF | 352.9 | 66.7 | 350.6 | 66.6 | 6 | 5.952 | 273.2 | 4.1 | 76.4 | 214.1 |
| BC048 | GD-AF | 351.9 | 69.5 | 349.3 | 69.3 | 6 | 5.931 | 72.5 | 7.9 | 72.7 | 218.9 |
| BC054 | GD-TH | 340.8 | 66.3 | 338.7 | 66.0 | 5 | 4.897 | 38.8 | 12.4 | 71.0 | 193.1 |
| BC060 | GD-TH | 7.3 | 67.3 | 4.9 | 67.4 | 5 | 4.897 | 38.7 | 12.5 | 76.7 | 254.8 |
| BC066 | IN-AF | 354.8 | 66.0 | 352.6 | 65.9 | 6 | 5.9 | 49.8 | 7.6 | 77.9 | 216.9 |
| BC073* | IN-AF | 6.0 | 67.1 | ... | ... | 4 | 3.867 | 22. | 19.8 | ... | ... |
| BC097 | GD-TH | 356.1 | 67.1 | 353.7 | 67.0 | 5 | 4.944 | 70.9 | 9.2 | 76.9 | 222.9 |
| BC103 | GD-AF | 358.1 | 68.7 | 355.5 | 68.6 | 5 | 4.952 | 85.0 | 8.5 | 75.1 | 230.3 |
| BC109 | GD-AF | 354.2 | 73.0 | 351.0 | 72.9 | 5 | 4.968 | 123.1 | 6.9 | 68.1 | 228.4 |
| BC115 | IN-AF | 34.1 | 66.2 | 339.0 | 65.9 | 4 | 3.982 | 168.2 | 7.1 | 71.3 | 193.1 |
| Mean field direction of Bear Creek sites* | | | | 351.0 | 69.1 | 11 | 10.968 | 310. | 2.6 | | |
| Mean VGP of Bear Creek sites* | | | | | | 11 | 10.919 | 123.6 | 4.12 | 73.3 | 222.0 |
| | | | | *Florence Lake* | | | | | | | |
| FL001 | IN-AF | 351.8 | 66.1 | 349.6 | 65.9 | 4 | 3.965 | 84.9 | 10.0 | 76.7 | 209.4 |
| FL007 | GD-AF | 339.1 | 67.1 | 336.9 | 66.7 | 4 | 3.990 | 289.7 | 5.4 | 69.5 | 194.0 |
| FL013 | IN-AF | 344.2 | 69.6 | 341.6 | 69.3 | 5 | 4.962 | 106.5 | 7.45 | 69.8 | 207.5 |
| FL019 | GD-AF | 337.1 | 73.6 | 344.0 | 73.2 | 4 | 3.979 | 140.1 | 7.8 | 66.1 | 220.4 |
| FL025 | GD-AF | 335.3 | 71.2 | 332.7 | 70.8 | 5 | 4.981 | 214.2 | 5.2 | 64.3 | 203.8 |
| FL031 | IN-AF | 342.3 | 65.4 | 340.2 | 65.1 | 5 | 4.985 | 275.1 | 4.6 | 72.4 | 191.2 |
| FL037 | GD-AF | 335.7 | 67.0 | 333.6 | 66.6 | 5 | 4.973 | 148.8 | 6.3 | 67.6 | 191.2 |
| FL043† | IN-AF | 339.1 | 64.1 | ... | ... | 3 | 2.980 | 18.2 | 29.8 | ... | ... |
| FL049 | IN-AF | 335.3 | 67.7 | 333.1 | 67.3 | 3 | 2.993 | 269.8 | 7.5 | 66.9 | 193.2 |
| FL055 | GD-AF | 329.5 | 67.7 | 327.4 | 67.2 | 5 | 4.969 | 128.5 | 6.8 | 63.5 | 190.0 |
| Mean field direction of Florence Lake sites† | | | | 337.6 | 68.1 | 9 | 8.984 | 491.2 | 2.33 | | |
| Mean VGP of Florence Lake sites† | | | | | | 9 | 8.960 | 201.6 | 3.64 | 68.8 | 199.8 |

Type: Lithology sampled and type of demagnetization used: GD, granodiorite; IN, mafic inclusions in granodiorite; TH, thermal demagnetization; af, alternating field demagnetization. In situ: field directions uncorrected for tilt. Corrected: field directions after tilt correction. Inc, inclination; Dec, declination; $N$, number of samples or sites averaged; $R$, vector sum; $\kappa$, precision parameter; $\alpha_{95}$, radius of 95% confidence in degrees ($A_{95}$ for poles). Locations used to compute VGPs are 37.35°N, 241.1°E for the Bear Creek sites, and 37.3°N and 241.0°E for the Florence Lake sites.

*BC073 is not included in means of sites due to a number of lightning damaged samples and discordant directions.

†FL043 is not included in means of sites due to lack of stable magnetization and to discordant directions.

nent plots [*Zijderveld*, 1967] displayed linear trends, suggesting the removal of a single component of magnetization. The magnetic directions were picked using a computer program modified after *Kirschvink* [1980], except for those samples whose magnetic signatures were too noisy for the computer algorithm; the magnetic directions for these samples were picked by hand from vector component plots. The natural remanent magnetization (NRM), as well as some of the initial demagnetization steps, were not used in determining the directions. The directions of the specimens cleaned by the two procedures were found to be essentially the same.

Most specimens exhibited a wide range of blocking temperatures. Averaged over 27 thermally demagnetized specimens from the Lake Edison pluton, the main component of magnetization was stable until 500°C with a standard deviation of 50°C. Five thermally demagnetized samples from the outcrop scale kinks showed an average blocking temperature of 321°C with a standard deviation of 21°C. The demagnetization history of two samples, which had stable magnetization to 565°C and 322°C, are shown in Figure 4. While the sample (bc012-00) that is deemed stable until 565°C does have a slight change in behavior at 363°C, the magnetic direction remains relatively stable until 565°C, especially when compared with the other sample (bc045-00).

## PALEOMAGNETIC RESULTS

### *Quadrangle Scale*

Approximately 80% of the samples exhibited a stable, single component of magnetization once the NRM was removed; this stable component was consistent within site. Most of the other 20% of the samples had NRM intensities an order of magnitude higher than the others within their sites, suggesting that these samples had been struck by lightning. The remaining samples exhibited either no stable magnetic direction or one that was widely inconsistent with their site means. Samples whose magnetic directions were more than 3 times the

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

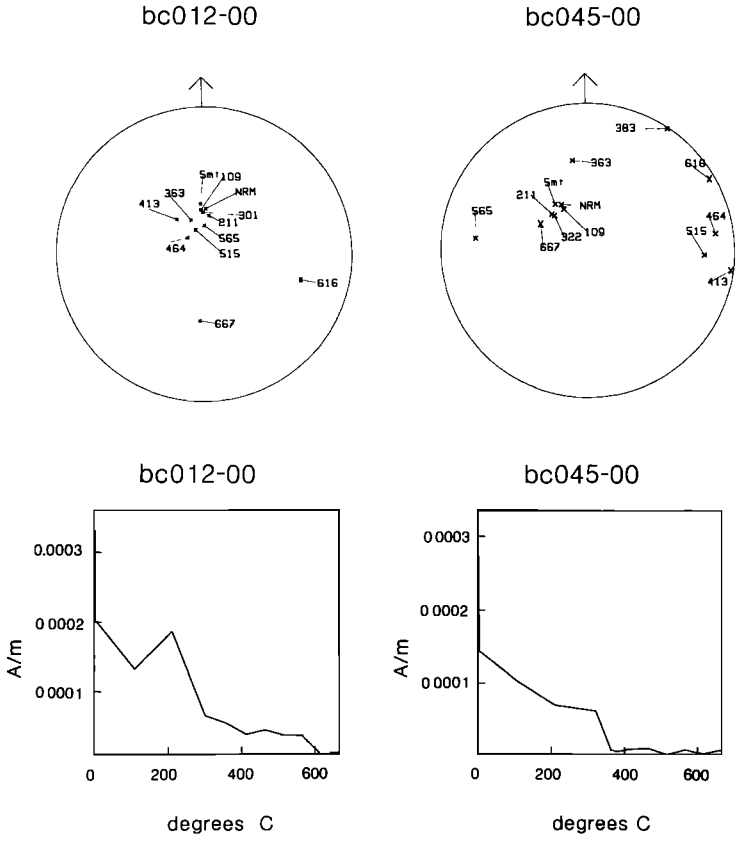



Fig. 4. Equal-area plots of the thermal demagnetization of two samples and associated graphs of magnetic intensity versus demagnetization temperature level. Each step past 5 mT of alternating field demagnetization is labeled in degrees Celcius. Small crosses indicate magnetic direction in the lower hemisphere, and large crosses indicate the upper hemisphere. Sample bc012-00 from the Bear Creek area has a stable magnetic direction until 565°C, while bc045-00 from the Trail Forks kink 2 is stable until 322°C.

95% confidence limit ($\alpha_{95}$) from the site mean were termed inconsistent and were excluded from further consideration.

The magnetic directions from the samples in each site were averaged using *Fisher* [1953] statistics to find site means (Table 1 and Figure 5). These site means were then averaged to find area means for the Florence Lake and Bear Creek areas (Figures 6 and 7, respectively). The mean directions from the two areas are compared in Figure 8. These site means and area means have been corrected for the 1° westward tilt of the Sierra Nevada batholith [*Frei et al.*, 1984].

Site pole positions were determined from the corrected declinations and inclinations using the locations listed in Table 1. Fisher statistics were used to determine mean poles from the site pole positions for both the Bear Creek and Florence Lake areas (Table 1). Figure 9 compares the paleomagnetic poles for the Bear Creek and Florence Lake areas with the pole deter-

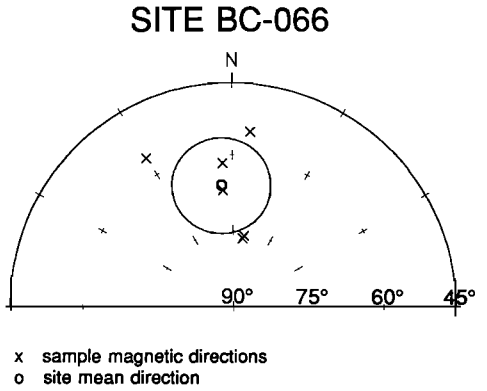



Fig. 5. Sample magnetic directions from site BC-066 and the site mean direction with its 95% confidence circle. The plotted area is the northern half of the lower hemisphere covering the range of plunges from 45° to 90° on an equal-area projection.

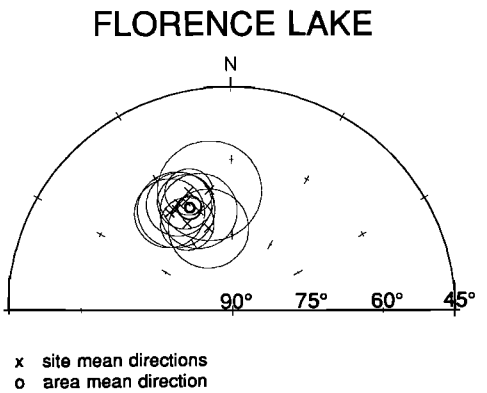



Fig. 6. Mean site directions for the Florence Lake sites and the area mean direction with its 95% confidence circle. See Figure 5 for a description of the projection.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

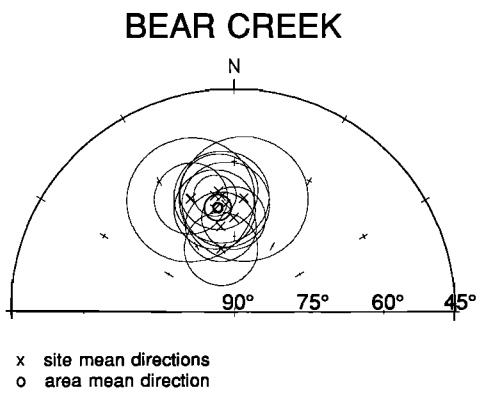



Fig. 7. Mean site directions for the Bear Creek sites and the area mean direction with its 95% confidence circle. See Figure 5 for a description of the projection.

mined from three plutons located 160 km to the north in the Sierra Nevada [*Frei*, 1985, 1986]. The two poles determined in this study fall within the confidence limit of the pole from the northern Sierra Nevada, and the pole for Florence Lake is closer to the paleomagnetic pole for the craton [*Irving and Irving*, 1982] than is the pole for the Bear Creek area.

Whether or not the Sierra Nevada batholith has undergone northward translation has been an important element of models developed to explain the formation of the west coast of North America [e.g., *Beck et al.*, 1981; *Beck*, 1986]. No northward translation of the Sierra Nevada is required by the present results. This is in agreement with other paleomagnetic results [*Gromme and Merrill*, 1965; *Gromme et al.*, 1967; *Kanter and McWilliams*, 1982; *Frei*, 1984]. The difference in paleomagnetic declination of the Bear Creek area with respect to the Florence Lake area and the associated 95% confidence limits were calculated from the declinations and mean poles for the two areas, using the method of *Cox* [1980], as modified by *Demarest* [1983]. This apparent rotation is 13.4° ± 7° clockwise.

*Outcrop Scale*

Since the outcrop scale kinks formed by folding [*Reches and Johnson*, 1976; *Collier*, 1978], a paleomagnetic study was made of them as a control experiment. The rotation of the inner limbs of the kink bands is easily measured at the out-

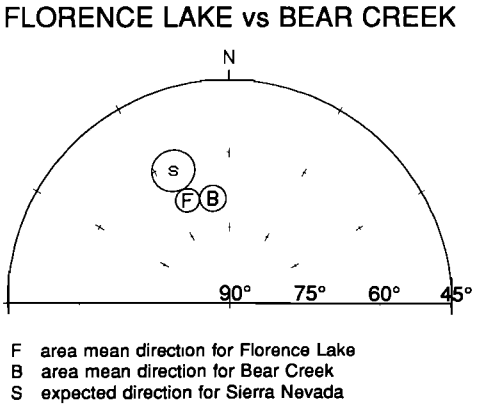



Fig. 8. Comparison of the area mean directions for Florence Lake and Bear Creek with their 95% confidence circles. The expected direction and its 95% confidence circle are also shown; these were determined from the pole for the Sierra Nevada at 90–100 Ma [*Frei*, 1985]. See Figure 5 for a description of the projection.

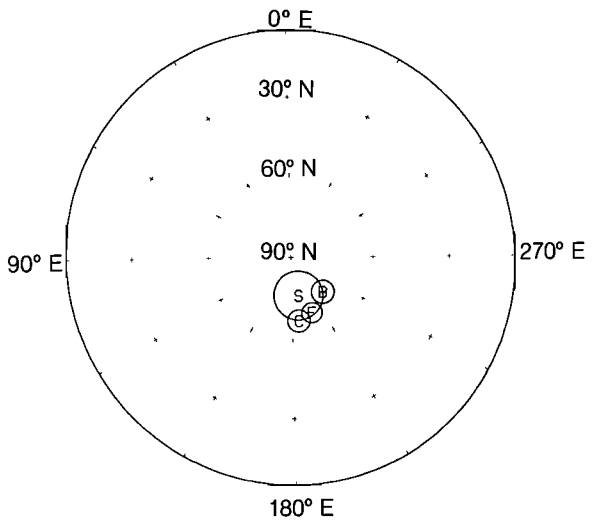



Fig. 9. An average pole position for the Sierra Nevada (S) at 90–100 Ma from *Frei* [1985] and its 95% confidence limit are compared with the pole positions determined for Florence Lake (F) and Bear Creek (B) and their 95% confidence limits. *Irving and Irving*'s [1982] pole for cratonal North America at 90 Ma (C) is also shown with its 95% confidence limits.

crop as the difference in strike with respect to the outer limbs (the inner limb of any sigmoid is the central section that connects the outer limbs (Figures 1 and 2)). This difference can be compared with the rotation calculated from the paleomagnetic data. For each of the three outcrop scale kink bands (the two at Trail Forks and the one at Kip Camp) the magnetic directions of the samples from the outer limbs (o's in Figure 3) were combined using Fisher statistics to find a mean value. The same was done for the inner limb samples (i's in Figure 3). The amounts of rotation, and the 95% confidence limits, between the inner and outer limbs were determined by the method of *McWilliams* [1984]. The method of *Demarest* [1983] after *Cox* [1980], which was used to calculate the rotation of the large-scale sigmoidal pattern, was not used for the small-scale kinks because the distance between the inner and outer limbs of the outcrop scale kinks is negligible, removing the need to transform the data into a coordinate system using the virtual geomagnetic poles. The amounts of rotation, as well as the fracture strikes of the inner and outer limbs, are presented in Table 2 and Figure 10.

In order to increase the sample size in the averages the data were combined in two additional ways. For the first the data from the two kinks at Trail Forks were merged such that the magnetic directions from the outer limbs were averaged, the magnetic directions from the inner limbs were averaged, and the rotation between the two averaged directions was calculated. For the second combination the data from all three kinks were merged, and a rotation was calculated between the combined inner limb and combined outer limb directions. The values for these combinations are also presented in Table 2, and the results for all kinks combined are shown in Figure 11. The physical rotation based on differences in strike ranges from 15° to 31° clockwise. All of the paleomagnetic rotations are smaller (−15° to 6°) than the difference in strike but have large 95% confidence limits; moreover, all but one of the observed paleomagnetic rotations are counterclockwise, while the sense of rotation predicted by the outcrop pattern is clockwise.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Paleomagnetic Results for Outcrop Kinks

| Site | Strike, deg | In Situ Dec, deg | In Situ Inc, deg | Corrected Dec, deg | Corrected Inc, deg | N | R | κ | $\alpha_{95}$ | r | Δr | Δ Strike |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | *Trail Forks 1* | | | | | | | |
| Outer limbs | 69 | 28.0 | 67.5 | 25.8 | 67.9 | 10 | 9.794 | 43.592 | 7.4 | | | |
| Inner limb | 100 | 20.9 | 70.2 | 18.2 | 70.5 | 2 | 1.991 | ... | ... | | | |
| Rotation | | | | | | | | | | −7.6 | 46.8 | 31 |
| | | | | | *Trail Forks 2* | | | | | | | |
| Outer limbs | 71 | 15.3 | 70.2 | 12.6 | 70.4 | 8 | 7.889 | 63.167 | 7.02 | | | |
| Inner limb | 100 | 20.1 | 64.4 | 18.1 | 64.7 | 4 | 3.863 | 21.880 | 20.1 | | | |
| Rotation | | | | | | | | | | 5.5 | 34.3 | 29 |
| | | | | | *Trail Forks: Combined* | | | | | | | |
| Outer limbs | 70 | 22.0 | 69.0 | 19.5 | 69.3 | 2 | 1.998 | ... | ... | | | |
| Inner limbs | 100 | 20.5 | 67.3 | 18.2 | 67.6 | 2 | 1.997 | ... | ... | | | |
| Rotation | | | | | | | | | | −1.3 | 19.1 | 30 |
| | | | | | *Kip Camp* | | | | | | | |
| Outer limbs | 85 | 23.7 | 65.0 | 21.7 | 65.3 | 5 | 4.902 | 40.875 | 12.11 | | | |
| Inner limb | 100 | 9.6 | 68.7 | 7.0 | 68.8 | 6 | 5.827 | 28.910 | 12.67 | | | |
| Rotation | | | | | | | | | | −14.7 | 38.9 | 15 |
| | | | | | *All: Combined* | | | | | | | |
| Outer limbs | 75 | 22.7 | 67.6 | 20.4 | 67.9 | 3 | 2.996 | 528.479 | 5.37 | | | |
| Inner limbs | 102 | 17.0 | 67.8 | 14.6 | 68.0 | 3 | 2.996 | 450.637 | 5.81 | | | |
| Rotation | | | | | | | | | | −5.8 | 14.0 | 27 |

Strike: average strike of joints. In situ: field directions uncorrected for tilt. Corrected: field directions after tilt correction. Inc, inclination; Dec, declination; $N$, number of samples or sites averaged; $R$, vector sum; $\kappa$, precision parameter; $\alpha_{95}$, radius of 95% confidence in degrees; $r$, rotation between outer limbs and inner limbs such that the geologically expected rotation is positive; $\Delta r$, 95% confidence limit for $r$ in degrees. Rotations and 95% confidence limits are calculated from corrected directions. Δ strike is the difference in strike between the inner and outer limbs.

While most of the outcrop-scale specimens were cleaned using af demagnetization, six of them were thermally demagnetized. For four of these six the magnetic direction was stable, on the average, until 321°C with a standard deviation of 21°C. The fifth was stable until 563°C, and the sixth had no stable magnetization. An example of a specimen (bc045-00) that was stable only to 322°C is shown in Figure 4.

## DISCUSSION

The 13° declination difference between the Bear Creek and Florence Lake areas is not enough to account for the 40°–60° difference in the strikes of photolineaments in these areas. Thus it is not possible that the sigmoidal lineament pattern formed fully by folding after the intrusions had cooled below their blocking temperatures. However, because the 95% confidence limit on the 13° discrepancy is only 7°, the apparent rotation is statistically significant and should be accounted for. While there are several possible mechanisms that could create a declination difference, this discussion will focus mostly on rotation because it is the only method that can form both the declination difference and the sigmoidal pattern of the photolineaments. By initially ignoring the other mechanisms, upper limits can be placed on the amount of rotation

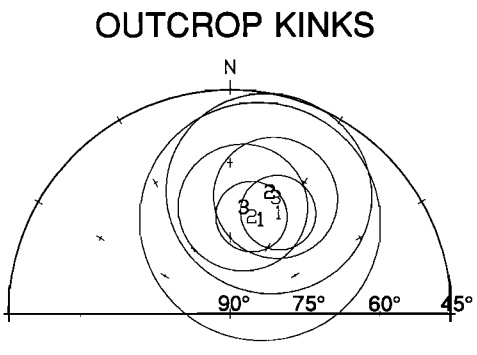



Fig. 10 Mean magnetic directions of the inner and the outer limbs of the three outcrop scale kinks with their 95% confidence circles. See Figure 5 for a description of the projection.

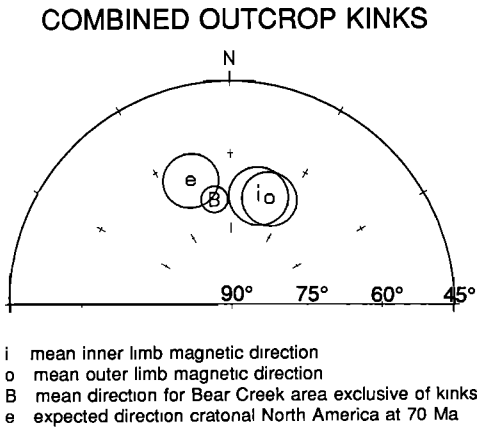



Fig. 11. Mean magnetic directions for the inner and the outer limbs averaged over the three outcrop scale kinks with their 95% confidence circles. Also shown, with their 95% confidence limits, are the expected direction and the direction for the rest of the Bear Creek area. The expected direction and its 95% confidence circle were determined from the pole for cratonal North America at 70 Ma [*Irving and Irving*, 1982]. See Figure 5 for a description of the projection.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

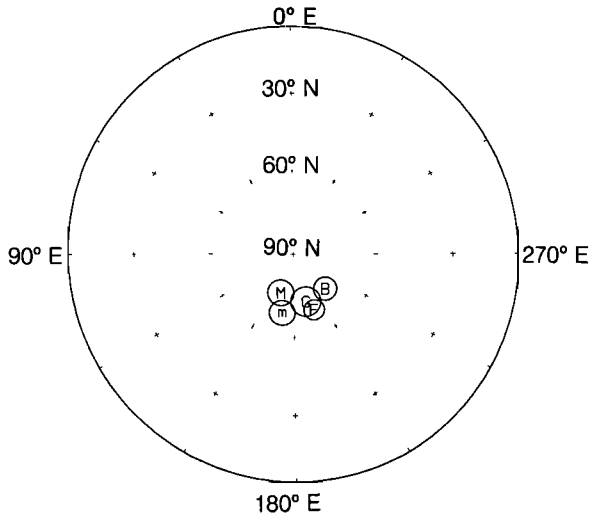



Fig. 12. Pole positions for Florence Lake (F), Bear Creek (B), and three poles from *Frei* [1985]: Grace Meadows pluton inclusion (G), Mount Gibson coarse grained (M), and Mount Gibson fine grained (m). Note that the Sierra Nevada pole for 90–100 Ma from *Frei* [1985] is not a simple average of the three poles from *Frei* [1985] shown here.

that took place after the rock mass cooled below the blocking temperature.

One possible mechanism that could have created the observed declination difference and the observed geology is that the sigmoidal pattern did form by kinking but that most of the kinking took place before the magnetization of the rock masses was set. It is also possible that the fractures formed initially with a 30°–50° kink and that the amount of kinking increased a small amount (13°) by folding. The possible scenarios include three factors: the shape of the fractures when they originally formed, the amount of folding that took place before the blocking temperature was reached, and the amount of folding that took place after the blocking temperature was reached. Given that the paleomagnetic data only provide control on the last factor and that this amount is much smaller than the geologically observed pattern, a large number of scenarios are possible that explain both the paleomagnetic data and the observed fracture pattern.

One possible mechanism that could have created the declination discrepancy, other than rotation, is tilting of the Bear Creek area with respect to the Florence Lake area. A small westward tilt of the Bear Creek area would bring its paleomagnetic direction into agreement with the direction obtained from the Florence Lake area.

If the two areas were magnetized at different times, a possible source of paleomagnetic declination discrepancy, other than movement of the rock masses while below their blocking temperatures, is apparent polar wander. Uncertainties in the relative ages of the Florence Lake and Bear Creek areas make it difficult to assess whether apparent polar wander could be the source of the observed declination discrepancy.

It is only possible to interpret the differences in paleomagnetic directions as relative tectonic movement if the differences are larger than the expected differences from nontectonic origins. Even though secular variation seems to have been very well averaged by the sites in this study (note the high $\kappa$ for the poles in Table 1) and these plutons cooled during a period of little apparent polar wander for North America [*Globerman*, 1986], it is still possible that the apparent 13° rotation is simply due to normal dispersion by an unknown mechanism among the poles from different parts of the Sierra Nevada. Figure 12 shows the two poles computed in this study along with three poles from other individual plutons in the Sierra Nevada [*Frei*, 1985, 1986]. These three poles did not come from regions with obvious geologic factors that suggest they should have different paleomagnetic poles, yet they have as much dispersion as the two poles from this paper. (Note that these three poles do not represent all of the data used to compute the average Sierra Nevada pole used from *Frei* [1986].)

Having ruled out the possibility that the large-scale sigmoidal photolineament pattern was formed by rotations of 40°–60° of the Bear Creek area (while the rocks were cooler than their blocking temperatures), the outcrop-scale kink patterns will now be discussed. Here there is geologic evidence that the sigmoidal fracture pattern did form by kinking [*Reches and Johnson*, 1976; *Collier*, 1978; *Davies and Pollard*, 1986]. Therefore, if the magnetization was acquired before kinking, the amount of rotation determined by paleomagnetic methods should be similar to the differnce in the fracture strikes between the inner and outer limbs of the kinks. This is approximately 30° clockwise for the Trail Forks kinks and 15° clockwise for the Kip Camp kink. For each of the individual kinks the expected rotation does fall within the 95% confidence limits of the apparent rotation determined by paleomagnetic methods (Table 2). However, this only reflects the large 95% confidence limits due to the small number of samples at each kink or site. It is noteworthy that for two of the three kinks the determined apparent rotation is in the opposite sense from the expected rotation. When any averaging of sites is made, the 95% confidence limit decreases, and the paleomagnetic results do not allow the kinks to have formed fully by folding after the blocking temperature was reached. While the paleomagnetic data do not preclude the geologically demonstrable rotation, they do not support it.

Comparison of the combined magnetic direction from the outer limbs of the kinks with the magnetic direction for the rest of Bear Creek (Figure 11) suggests that the entire kink set is rotated. However, the fracture strikes of the outer limbs of the kinks are the same as those found throughout the rest of the Bear Creek area; this suggests that the kink sets are not rotated with respect to the rest of the Bear Creek area. One possible explanation is that there was a reheating event that reset the magnetic directions of the samples from the outcrop-scale kinks. There is some circumstantial evidence for this; the magnetic directions of the samples from the sites devoid of small-scale kinks are stable until 500° ± 50°C, whereas those at the kinks are stable until only 321° ± 21°C. The one sample from a kink site with a high blocking temperature (563°C) is from the outer limb of the kink, where the least amount of deformation occurred. Assuming that the hydrothermal fluids depositing these minerals also were at about 360°C (A. Kilinc, unpublished data, 1983), there is a striking agreement between this result and the blocking temperature of the magnetization. It suggests that the kinks were formed prior to, or contemporaneous with, a reheating event. However, while this evidence allows such a reheating event, it does not require it.

In any case, the difference between the magnetic directions of the outer limbs of the kinks and that of the rest of Bear

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Creek cannot be explained by heating and remagnetization of the kinks due to joint filling emplacement. Samples from the joint fillings have been dated at 71 and 75 Ma; the magnetic direction of the kinks is closer to the Bear Creek magnetic direction than the expected direction [*Irving and Irving*, 1982] for this time (Figure 11). If a reheating event did take place, then the outer limbs of the kinks should have a magnetic direction closer to the expected direction for this time.

The difference in declination of the outer limbs of the kinks with respect to the expected direction is 38.4° ± 20.9 using the method of *McWilliams* [1984]. Since the joints in the outer limbs of the kink sets did not rotate with respect to the rest of the Bear Creek area, while the magnetic directions of the samples between the joints in the outer limbs and the rest of the Bear Creek area do display a discrepancy in declination, it is possible that there was continuous deformation of the rock mass between the joints. This might be a result of the kinking process proposed by *Davies and Pollard* [1986].

It is also interesting that this reheating to 360°C due to the emplacement of the joint fillings is not apparent in the blocking temperatures away from the small-scale kinks, despite the presence of faults with mineral-filled cracks throughout the region. However, the fault spacing in the small-scale kinks is of the order of 30 cm, while the fault spacing away from the kinks is considerably greater. The samples taken outside the small-scale kinks were taken as far as possible from any faults. If the source of reheating is the hydrothermal activity in the cracks associated with faulting and kinking, this greater spacing would contribute to these samples not being reheated. Thus the small fault spacing may be necessary to uniformly heat the rock mass by the hydrothermal activity in the fractures. This uniform heating and small fault spacing may have been causally related to the small-scale kinking.

## CONCLUSIONS

The detailed study of the outcrop scale kink bands by *Davies and Pollard* [1986] demonstrates that the bands formed due to a rotation of the inner section with respect to the outer limbs. They suggest that the quadrangle-scale sigmoidal photo lineament bands might reflect an analogous process during which the Bear Creek area would have rotated 40°–60° clockwise with respect to the Florence Lake area. The quadrangle scale mapping, and detailed mapping of the left-lateral fault zones (S. Martel, personal communication, 1986), in the Mount Abbot quadrangle suggests that such a large rotation has not occurred at the quadrangle scale. The results of this paleomagnetic investigation indicate there is a 13.4° ± 7° difference in paleomagnetic declination between the Bear Creek and Florence Lake areas. Even if this discrepancy in declination was caused by a rotation of the Bear Creek area with respect to the Florence Lake area, the paleomagnetic data show that the sigmoidal photo lineament pattern did not form completely by rotation after the rock mass had cooled below the blocking temperature. Thus, if rotation is the mechanism that caused the quadrangle scale photolineament pattern to be sigmoidal, the rotation must have largely occurred at temperatures greater than 500° ± 50°C. An obvious alternative is that the sigmoidal pattern was not caused by rotation and the declination difference is due to tilting, apparent polar wander, or paleomagnetic dispersion.

At the small-scale kink sites the paleomagnetic data indicate that although no rotation is evident, a reheating event (due to the circulation of hydrothermal fluids) which removed the rotation from the paleomagnetic record could have occurred. This possible reheating event is not apparent away from the outcrop scale kinks, suggesting that the reheating event could be causally related to the outcrop-scale kinking. The data further suggest that there may have been continuous deformation during the formation of the kinks. Another result of this study is further paleomagnetic evidence that the Sierra Nevada pluton has not moved significantly northward with respect to the North American Craton.

*Acknowledgments.* I am grateful to my advisor, Allan Cox, for his guidance. Dave Pollard and Leah Frei suggested this project, and Leah gave many suggestions in the early stages of it. Steve Martel helped me understand the geologic history of this area. I thank my drilling crew Brian Hart, Les Nakae, Andy Michael, Paul Layer, and Jim Beaujon. Critical reviews by Doug Wilson, Myrl Beck, Tracy Vallier, an anonymous reviewer, and the associate editor helped improve this manuscript.

## REFERENCES

Beck, M. E., Jr., Model for Late Mesozoic–Early Tertiary tectonics of coastal California and western Mexico and speculations on the origin of the San Andreas fault, *Tectonics*, *5*, 49–64, 1986.

Beck, M. E., Jr., R. F. Burmester, D. C. Engebretson, and R. Schoonover, Northward translation of Mesozoic batholiths, western North America: Paleomagnetic evidence and tectonic significance, *Geofis. Int.*, *20*, 143–162, 1981.

Chen, J. H., and J. G. Moore, Uranium-Lead isotopic ages from the Sierra Nevada batholith, California, *J. Geophys. Res.*, *87*, 4761–4784, 1982.

Collier, M., Ultimate locking angles for conjugate and monoclinal kink bands, *Tectonophysics*, *48*, T1–T6, 1978.

Cox, A., Rotation of microplates in western North America, The Continental Crust and Its Mineral Deposits, edited by D. W. Strangway, *20*, *Spec. Pap. Geol. Assoc. Can.*, 305–321, 1980.

Davies, R. K., and D. D. Pollard, Relations between left-lateral strike-slip faults and right-lateral monoclinal kink bands in granodiorite, Mt. Abbot quadrangle, Sierra Nevada, California, *Pure Appl. Geophys.*, *124*, 177–201, 1986.

Demarest, H. H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. Geophys. Res.*, *88*, 4321–4328, 1983.

Evernden, J. F., and R. W. Kistler, Chronology of emplacement of Mesozoic batholithic complexes in California and western Nevada, *U.S. Geol. Surv. Prof. Pap.*, *623*, 42 pp., 1970.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295–305, 1953.

Frei, L. S., Paleomagnetic constraints on extension in the Basin and Range and in the North Atlantic area, Ph.D. dissertation, Stanford Univ., Stanford, Calif., 1985.

Frei, L. S., Additional paleomagnetic results from the Sierra Nevada: Further constraints on Basin and Range extension and northward displacement in the western United States, *Geol. Soc. Am. Bull.*, *97*, 840–849, 1986.

Frei, L. S., J. R. Magill, and A. Cox, Paleomagnetic results from the central Sierra Nevada: Constraints on reconstructions of the western United States, *Tectonics*, *3*, 499–537, 1984.

Globerman, B. R., Paleomagnetic reinvestigation of Middle Cretaceous alkalic igneous intrusive rocks from central Arkansas (abstract), *Eos Trans. AGU*, *67*, 922, 1986.

Gromme, C. S., and R. T. Merrill, Paleomagnetism of Late Cretaceous granitic plutons in the Sierra Nevada, California: Further results, *J. Geophys. Res.*, *70*, 3407–3420, 1965.

Gromme, C. S., R. T. Merrill, and J. Verhoogen, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California, and its significance for polar wandering and continental drift, *J. Geophys. Res.*, *72*, 5661–5684, 1967.

Hamilton, W., and W. B. Meyers, Cenozoic tectonics of the western United States, *Rev. Geophys.*, *4*, 509–549, 1966.

Irving, E., and G. A. Irving, Apparent polar wander paths Carbonifer-

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ous through the Cenozoic and the assembly of Gondwana, *Geophys. Surv.*, *5*, 141–188, 1982.

Kanter, L. R., and M O. McWilliams, Rotation of the southernmost Sierra Nevada, California, *J. Geophys. Res.*, *87*, 3819–3830, 1982.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

Kistler, R. W., P. C. Bateman, and W. W. Brannock, Isotopic ages of minerals from granitic rocks of the Central Sierra Nevada and Inyo Mountains, California, *Geol. Soc. Am. Bull.*, *76*, 155–164, 1965.

Lockwood, J. P., and P. A. Lydon, Geologic map of the Mount Abbot quadrangle, Central Sierra Nevada, California, *Map GQ-1155*, U S. Geol. Surv., Reston, Va., 1975.

Lockwood, J. P., and J. G. Moore, Regional deformation of the Sierra Nevada, California, on conjugate microfault sets, *J. Geophys. Res.*, *84*, 6041–6049, 1979.

McWilliams, M., Confidence limits on net tectonic rotation, *Geophys. Res. Lett.*, *11*, 825–827, 1984.

Reches, Z., and A. M. Johnson, Asymetric folding and monoclinal kinking, *Tectonophysics*, *35*, 295–334, 1976.

Segall, P., and D. D. Pollard, Nucleation and growth of strike slip faults in granite, *J. Geophys. Res.*, *88*, 555–568, 1983*a*.

Segall, P., and D. D. Pollard, Joint formation in granitic rock of the Sierra Nevada, *Geol. Soc. Am. Bull.*, *94*, 563–595, 1983*b*.

Segall, P., and D. D. Pollard, From joints and faults to photolineaments, in *Proceedings of the 4th International Conference on Basement Tectonics*, edited by I. Ramberg and R. H. Gabrielsen, pp. 11–20, 1984.

Stern, T. W., P. C. Bateman, B. A. Morgan, M. F. Newell, and D. L. Peck, Isotopic U-Pb ages of zircon from the granitoids of the central Sierra Nevada, California, *U.S. Geol. Surv. Prof. Pap.*, *1185*, 17 pp., 1981.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism, Dev. Solid Earth Geophys.*, vol. 3, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, Amsterdam, 1967.

S. L. Ross, U.S. Geological Survey, Branch of Pacific Marine Geology, 345 Middlefield Road, MS 999, Menlo Park, CA 94025.

(Received June 11, 1987;
revised January 18, 1988;
accepted February 22, 1988.)

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11711 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License